



中国科学院大学
University of Chinese Academy of Sciences

Detection of mild depression in adolescents based on multimodal data from electroencephalogram and eye movement tracking

基于脑电图与眼动追踪多模态数据检测青少年轻度抑郁症

小组成员：舒XX、吴XX、赵XX、Illusionna

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01

Research background



A Silent Struggle: How Do We See the Unseen? (一场无声的挣扎：我们如何看见看不见的痛苦？)

Depression affects hundreds of millions, yet it often remains invisible.
Why is it so hard to detect accurately?



The Necessity of Research on the Incidence of Adolescent Depression:



- **Large Patient Base And Rising Trend :** Global and Chinese data indicate that the detection rate of adolescent depression **increases significantly** with grade level.
- **Severe Burden And Profound Harm:** Depression is the leading cause of disability in adolescents, easily triggering a **sharp decline in academic performance and social withdrawal** even lead to self-harm and suicide.
- **Large Service Gap And Low Recognition Rate:** Clinical misdiagnosis and underdiagnosis are common. Most depressed adolescents do not receive standardized intervention. Beyond that, depression in adolescents is **often overlooked** in comparison to adults.



A Widespread Mental Health Crisis

According to the WHO, **depression** is a leading cause of disability worldwide, affecting an estimated **280 million** people globally (as of 2023).



Mild Depression: A Hidden Epidemic

Often undiagnosed, mild symptoms significantly impair daily functioning and productivity, acting as a precursor to more severe depressive episodes.



Limitations of Traditional Diagnostic

Such as scale assessments and clinical interviews, rely heavily on **subjective judgment** and are prone to **misdiagnosis and underdiagnosis**.



Core Concepts

- **Eye-Tracking Technology 眼动追踪**: Non-invasive technology, reflects human cognitive and emotional processing, its features can reflect attentional biases in depressed patients. (Focuses on external cognitive behavior)
- **EEG (Electroencephalogram) 脑电图**: Non-invasive technology, records brain electrical activity, serves as a reliable neural biomarker for depression detection, with higher sampling frequency. (Focuses on internal brain neural activity)

Our research mainly focus on **mild depression**.

Mild depression refers to a less severe form of depression marked by persistent low mood over a long period.

It mostly **occurs in adolescents** and is often **overlooked**, even though it tends to relapse easily and needs early detection. In this study, it is specifically defined as a score between **32 and 38 on the DSM-V (量表体系)** depression scale for young people **aged 11–17**, and it can be recognized through unusual brain wave (EEG) and eye movement patterns.



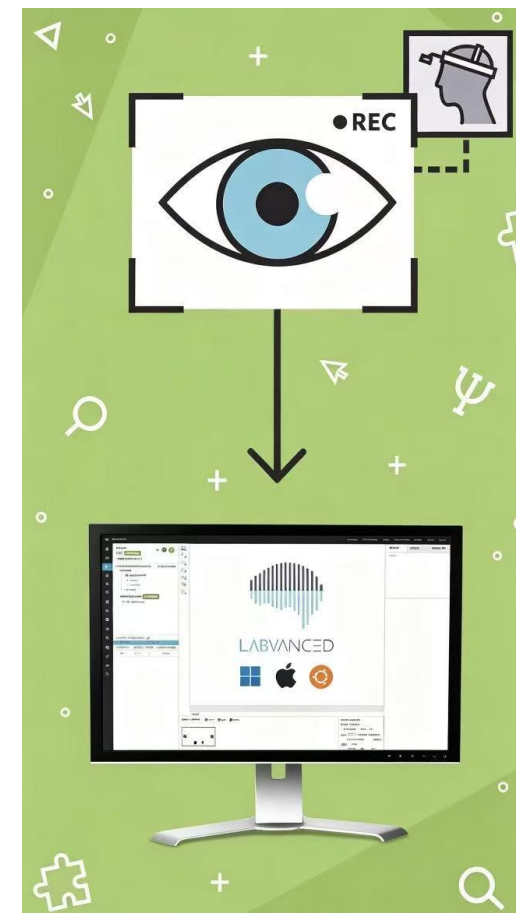
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Research Progress



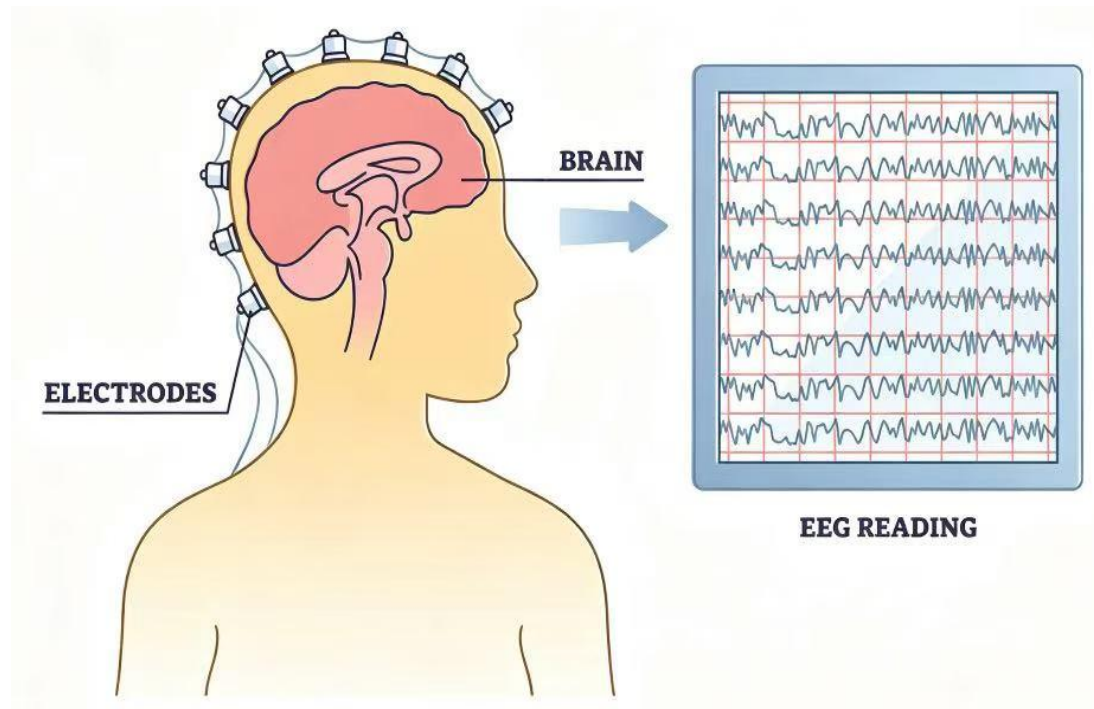
Part 1: Eye tracking in Depression Detection

Source	Methods	Results and conclusions	Limitations
Alhowinem et al. (2013)	First linked eye movement indicators with depression detection	Eye metrics reflect depressive attentional bias, building the foundation for application	Qualitative analysis only, no deep learning
Stolicyn et al. (2022)	Developed a depression assessment system	Extracted facial and eye-tracking features, 79% accuracy via SVM (classic machine learning model)	No mild depression classification, limited generalization
Yang et al. (2022)	Proposed a time-space eye pattern analysis method	Detected depressive visual attention patterns, 76.25% accuracy in distinguishing depressed from healthy subjects	Small sample size, low sensitivity to mild depression





Part 2: EEG Research in Depression Detection



- ① Mahato & Paul (2020) classified depressed and healthy subjects using EEG band features and SVM, achieving 88.33% accuracy.
 - Limitation: Traditional machine learning, manual feature extraction



Deep Learning

Deep learning extracts features automatically and effectively.

- ② Yin et al. (2023) proposed a Transformer-CNN hybrid model, which outperforms previous methods.
- ③ Tigga & Garg (2023) introduced an attention-based Transformer model, achieving 98.67% accuracy in depression detection.



Multi-modal Data Fusion

- ④ Ding et al. (2019) classified **major** depressive disorder via EEG, eye-tracking and skin electrical signals, with 79.63% classification accuracy.
- Limitation: Still in the initial stage, with insufficient research on **mild** depression.

Lack of efficient fusion strategies for EEG and eye-tracking in mild depression detection !



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Materials and methods



I: Subject Information

group	number of people	average age	inclusion criteria
Mild to moderate depression	21	16.5	DSM-V score: 32 - 38
health comparison	28	17.5	No depressive symptoms

II: Experimental paradigm and equipment

Driving scenario: 20 trials, including sudden appearance of pedestrians and vehicle collisions

EEG equipment

Eye-tracking device

III: Preprocessing and Feature Extraction

modal	feature extract extraction
EEG	statistical characteristic、Hjorth、nonlinear character
Eye-tracking	AOI sample percentage, AOI fixation time, fixation distribution



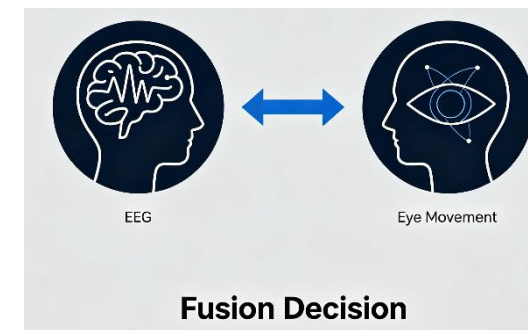
Traditional machine learning



EEG features
(carrot chunks)
+ Eye movement features
(lettuce leaves)
= Mixed salad

VS

Deep learning fusion - MTNet

**Disadvantages:**

The features extracted manually may not be comprehensive enough.

Simple concatenation ignores the deep interrelationship between the two types of data.

Advantages:

Automatically extract the most effective features

It is possible to learn the complex relationships between two types of data.

The effect has been significantly improved.

The traditional method is like "simply cutting the ingredients and mixing them together", while our MTNet is like "asking two experts to analyze separately and then discuss together". Therefore, it can discover more detailed and specific information related to depression.



Early vs. Middle vs. Late

Early fusion: Input layer directly concatenates raw features → FCNN

Intermediate fusion (adopted by MTNet): Features of each modality are extracted separately and then fused in the hidden layer.

Late fusion: Each modality is independently modeled, and the prediction results are weighted averaged (average weighted / entropy weighted)

Integration stage	accuracy rate	sensitivity	specificity	F1 score
Early integration	85.91%	85.00%	82.76%	0.837
Intermediate integration	91.79%	90.48%	92.86%	0.904
Late fusion	89.71%	90.00%	89.66%	0.878



- Intermediate fusion of the best: Capable of effectively adapting to the heterogeneity of multimodal data and capturing the interactions between modalities
- **MTNet + Intermediate Fusion** achieves the best detection performance (91.79%), which is superior to single-modal and feature-level fusion.



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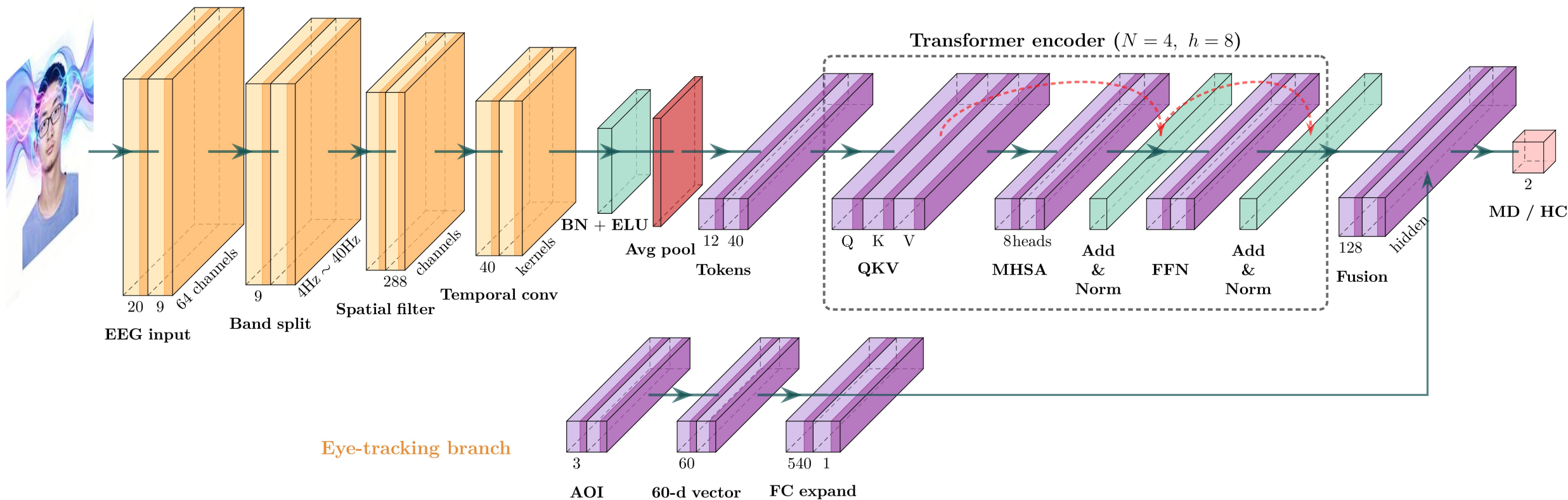
Analysis and Conclusion



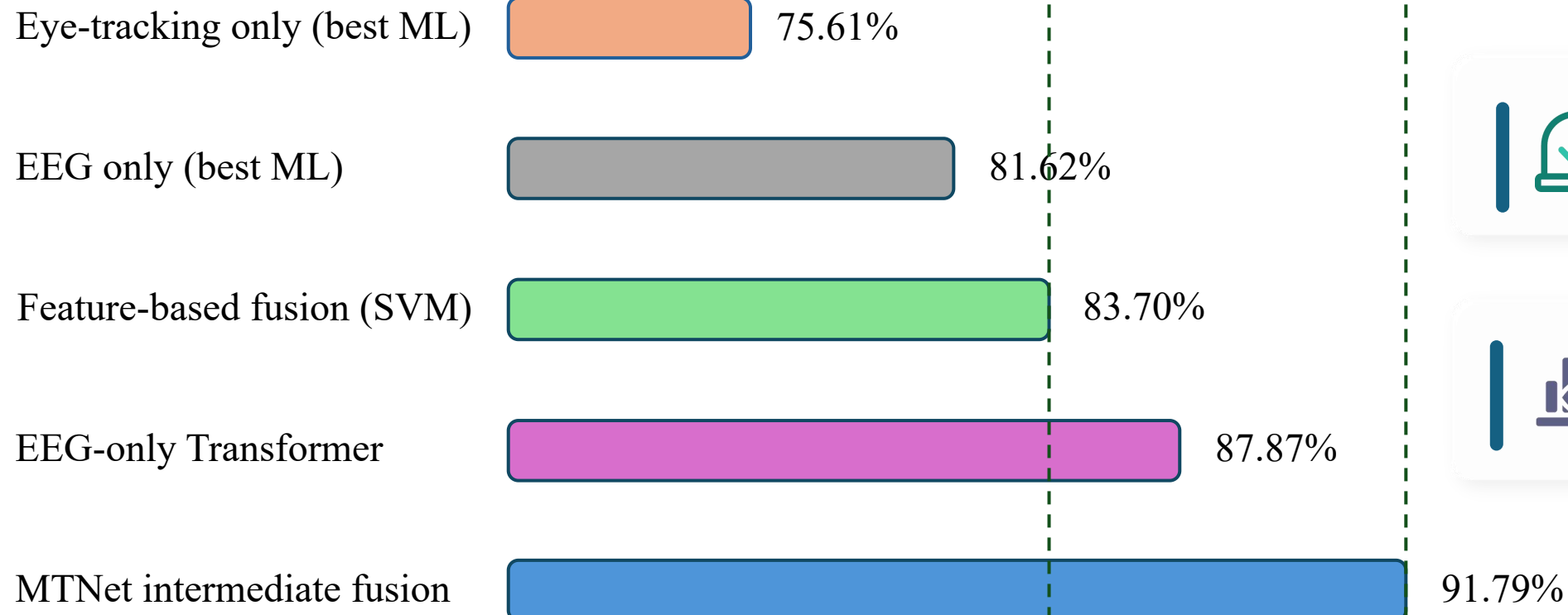
MTNet — Multimodal Transformer Architecture

EEG and eye-tracking streams are encoded separately, then fused in a 128-node hidden layer

EEG branch



Target object: 49 adolescents; mild depression: DSM-V youth scale 32-38; reported MTNet; accuracy 91.79%, F1 score 0.904.



+8.09% over feature-based fusion

Key readout



Best accuracy
91.79%



F1-score peak
0.904

Signal story: **eye behavior + EEG neural activity** > Transformer-based intermediate fusion > stronger screening performance



Error Analysis: Why Is MTNet Better for Screening?

The intermediate fusion setting **cuts total mistakes in half** and reduces missed mild-depression cases.

Feature-based

24
Correct HC

4
False alarm

4
Missed MD

17
Correct MD

Total errors: 8 Missed MD: 4

Early

25
Correct HC

3
False alarm

4
Missed MD

17
Correct MD

Total errors: 7 Missed MD: 4

Intermediate

26
Correct HC

2
False alarm

2
Missed MD

19
Correct MD

Total errors: 4 Missed MD: 2

Late

26
Correct HC

2
False alarm

3
Missed MD

18
Correct MD

Total errors: 5 Missed MD: 3

- Intermediate fusion significantly reduces misclassification, cutting total errors from 8 to 4 and missed mild-depression cases from 4 to 2.
- MTNet provides a more balanced screening result, achieving high sensitivity and specificity while reducing the risk of missed depression detection.



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References and About us



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Zhao Poling

Control the process
Clearly assign tasks to team members
Complete the research methods section

Shu Anqi

Complete the research background section
Provide the presentation outline

Wu Qirui

Complete the research progress section
Provide suggestions for revisions

Ye Binghui

Complete the data analysis and result section
Provide professional guidance.





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University of Chinese Academy of Sciences

Thank you for listening!

